

1. Prevalence of Odontogenic Neoplasms in Educational Hospitals and Institutions in Cairo.

(Cross Sectional Study)

**إنتشار الأورام سنية المنشأ في المستشفيات والمؤسسات التعليمية بالقاهرة.
(دراسة مقطعية)**

Submitted to the Faculty of Dentistry, Cairo University in partial fulfillment of the requirements for PH.D Degree in Oral and Maxillofacial Pathology

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Protocol Checklist				
Section and topic	Item No.	Checked item	Reported on page NO.	Reviewer's check
<u>I. Administrative information</u>				
	1	Title: PECOS elements included	1	
	2	Protocol registration	1	
	3	Protocol version	1	
	4	Funding	1	
	5	Roles and responsibilities of the authors	1	
<u>II. Introduction</u>				
	6a	Scientific background	2	
	6b	Specific objectives	3	
<u>III. Methods</u>				
A) Study design and settings	7	Study design	4	
	8	Settings	4	
B) Participants	9a	Eligibility criteria	4	
	10	Inclusion and Exclusion criteria	4	
C) Variables	11	Clearly define all variables including outcomes and exposures	5	
	12	For each variable of interest, give data sources/ measurement	5	
	13	Addressing potential sources of bias	5	
D) Study size	14	Study size	6	
E) Statistical methods	15	Statistical methods	6	

<u>IV- References</u>		References by reference manager	7	
<u>Evidence based committee (Reviewers)</u>				
Name		Signature	Date	
1.				
2.				
<u>Research plan committee</u>				
Name		Signature	Date	
1.				
2.				

I. Administrative information:

1. Title:

Prevalence of Odontogenic Neoplasms in Educational Hospitals and Institutions in Cairo.

2. Protocol registration

Cairo University,

3. Protocol version:

Version (1)

4. Funding:

Self-funding.

5. Roles and responsibilities:

a- The Principle Supervisor:

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c- The Investigators:

- **Dr. Eman Desouky Taha**, Lecturer of epidemiology and biostatistics, National Cancer Institute, Cairo University.
- **Leena Ali Mohamed** (The Principle Investigator).

II. Introduction:

6a. Odontogenic tumors (OTs) comprise a complex group of lesions with diverse clinical behaviors and histological types. Some of these lesions are true neoplasms while others may represent malformations similar to tumors (hamartomas) **(Luppi et al., 2018)**. Odontogenic tumors are relatively rare lesions, accounting for about 2% of all maxillofacial clinicopathological specimens sent for diagnosis to pathology services **(Rubini et al., 2017)**.

The knowledge of the epidemiological characteristics such as anatomic location, age, and gender of the patients with odontogenic tumors is extremely valuable in the development of differential diagnoses **(Luppi et al., 2018)**. Since a number of tumours of the jaws share the same clinical and radiographic features, **(Jones et al., 2006)** the diagnosis of odontogenic tumours ordinarily requires a detailed analysis of clinical, radiographical, and histopathological findings **(Johnson et al., 2014)**. Despite some sampling bias, diagnostic services presently, are the most reliable source of information regarding the relative frequency and clinicopathologic features of odontogenic tumours **(Shekar et al., 2014)**.

Epidemiological studies are crucial because they allow us to establish more precisely the occurrence of OTs in different populations, which in turn helps in making of a provisional diagnosis and further planning of the biopsy based on the clinical and radiographic features. It also aids in patient counseling and scheduling of treatment **(Sekerici et al., 2015)**.

A lot of the epidemiological researches relating to odontogenic tumours were conducted in other countries **(Kebede et al., 2017; Silva et al., 2017; Suhasini Palakshappa et al., 2017)**, despite the large number of studies on Odontogenic cysts, studies on OT are relatively rare **(Soluk-Tekkesin and Wright, 2017)**. Hence the aim of the present study is to assess the relative prevalence of OTs in educational hospitals and institutions in Cairo. Also efforts will be made to correlate the observations of current study with the previous published studies.

6b. PECO

Patients: Patients reports with odontogenic neoplasms in Cairo's educational hospitals and institutions.

Exposure: Odontogenic neoplasms diagnosed based on histopathological examinations.

Control: Not applicable.

Outcome: Prevalence rate.

	Outcome name	Measuring unit
Primary outcome	Prevalence rate of odontogenic neoplasms.	$\frac{\text{Number of odontogenic neoplasms cases}}{\text{Total no. of all cases}}$
Secondary outcome	Prevalence of odontogenic neoplasms in relation to: • Age (all ages) • Gender (male & female). • Site affection.	---

Research Question:

- What is the prevalence of odontogenic neoplasms in educational hospitals and institutions in Cairo?

Objective

- The objective of this study is to determine the prevalence of odontogenic neoplasms diagnosed histopathologically over a period of 5 years (from 2014 to 2018) in Cairo's educational hospitals and institutions. Furthermore, stratification of age, gender and site affection of each individual lesion will be performed.

III. Methods:

7. Study design

Retrospective cross-sectional study.

8. Settings

1. In Cairo city.
2. The date of data collection of odontogenic tumors reports (from 2014 to 2018).
3. The data will be collected from archive of all Cairo's educational hospitals and institutions.

A) Participants

9. Eligibility criteria and selection methods

Patient adhering to all following criteria will be included:

Inclusion criteria:

- ✓ Age: all age groups.
- ✓ Gender: both genders.
- ✓ Site: maxilla and mandible.
- ✓ All odontogenic neoplasms that fit the histological classification of the World Health Organization (2017) will be included in the study.
- ✓ All reports should include in period between (2014 and 2018).

Exclusion criteria:

- ☒ Any patient's report that doesn't record in archive of educational hospitals and institutions in Cairo.
- ☒ Any reports not found in period between (2014 and 2018).
- ☒ Non-odontogenic lesions.

10. Matching criteria and allocation ratio

Not applicable.

B) Variables

11. Details about variables

	Outcome name	Measuring unit
Primary outcome	Prevalence rate of odontogenic neoplasms.	$\frac{\text{Number of odontogenic neoplasms cases}}{\text{Total no. of all cases}}$
Secondary outcome	Prevalence of odontogenic neoplasms in relation to: • Age (all ages) • Gender (male & female). • Site affection.	---

12. Data sources and management

The data will be collected from all Cairo's educational hospitals and institutions in a period between (2014 and 2018). This data will be analyzed according to age, gender and site affection with the different types of odontogenic tumor according to the histological classification of the World Health Organization (2017) (Soluk-Tekkeşin and Wright, 2017).

13. Addressing potential sources of bias

- Improper diagnostic methods.
- Poor investigations for some difficult cases.

D) Study size

14. Study size

No sample size could be applied on our epidemiological study.

E) Statistical methods

15. Statistical methods

Categorical variables will be expressed in proportions or percentages; Chi square and Fisher's exact tests will be applied as appropriate. Continuous variables will be examined for normality using Kolmogorov-Smirnov test and Shapiro-Wilk test. They will be expressed using range, mean, median $\pm$ standard deviation (normal variables); range, median and interquartile range (non-normal variables). Test of significance will be used for comparison/association. The results were considered statistically significant for a value of $p \leq 0.05$.

IV- References:

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